



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 11/09/2025*

EXERCISE

It is given an Auto-Regressive (AR) process $S[n]$ of order 1, $S[n] = \rho S[n-1] + I[n]$, where $I[n]$ is the zero-mean innovation process with variance σ_I^2 .

(1) Derive the system function $L(z)$ of the innovation filter, its zeros and poles, and determine when it is a low-pass, high-pass, or band-pass filter, according to the value of parameter ρ . Derive the power of the process σ_s^2 , the autocorrelation function (ACF) $R_s[m] = E\{S[n]S[n+m]\}$ and the power spectral density (PSD) $S_s(e^{j2\pi f}) = FT\{R_s[m]\}$ of the process $S[n]$ and plot them for $\rho = 0.9$ and $\sigma_I^2 = 2$.

(2) Derive the optimal linear predictor of order $P=2$ of the process $S[n]$, i.e. the linear minimum mean square error (LMMSE) estimator of $S[n]$ given the observed data $S[n-1]$ and $S[n-2]$. Derive its mean square error (MSE) $\sigma_\varepsilon^2 \triangleq E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])^2\}$. Calculate numerically the expression of the estimator and its MSE when $\rho = 0.9$ and $\sigma_I^2 = 2$.

(3) Assuming that the signal of interest $S[n]$ is observed in the presence of additive white Gaussian noise (AWGN), i.e. $X[n] = S[n] + W[n]$, where $W[n]$ is AWGN with variance σ_w^2 , derive the optimal linear predictor of order $P=2$ of the process $S[n]$, i.e. the LMMSE estimator of $S[n]$ given the observed data $X[n-1]$ and $X[n-2]$ (in this case we talk about filtering and prediction) as a function of ρ and of the Signal to Noise Ratio (SNR) $\gamma = \sigma_s^2 / \sigma_w^2$. Derive its MSE. Calculate numerically the expression of the estimator and its MSE when $\rho = 0.9$, $\sigma_I^2 = 2$, and $\sigma_w^2 = 5$, compare the MSE with the one previously derived.



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Solution.

$$(1) S[n] = \rho S[n-1] + I[n]$$

$$S(z) = \rho z^{-1} S(z) + I(z) \Rightarrow L(z) = \frac{S(z)}{I(z)} = \frac{1}{1 - \rho z^{-1}} = \frac{z}{z - \rho}$$

The innovation filter has 1 zero at $z = 0$ and 1 pole at $z = \rho$. Hence, the innovation filter is a high-pass filter if $\rho < 0$ and a low-pass filter if $\rho > 0$, and so the process.

The ACF and the PSD are given by (see the course vugraphs for the derivation):

$$R_s[m] = E\{S[n]S[n+m]\} = \frac{\sigma_I^2}{1 - \rho^2} \rho^{|m|}, \text{ hence } \sigma_s^2 = R_s[0] = \frac{\sigma_I^2}{1 - \rho^2}.$$

$$S_s(e^{j2\pi f}) = \sigma_I^2 \left| L(e^{j2\pi f}) \right|^2 = \sigma_I^2 \left| \frac{1}{1 - \rho e^{-j2\pi f}} \right|^2 = \frac{\sigma_I^2}{1 + \rho^2 - 2\rho \cos(2\pi f)}$$

When $\rho = 0.9$, $S[n]$ is a low-pass process.

$$(2) \hat{S}[n] = h[1]S[n-1] + h[2]S[n-2] + b$$

Since $S[n]$ is a zero mean process we have that: $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.

$$E\{\varepsilon[n]S[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - \hat{S}[n])S[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])S[n-m]\} = 0, \quad m = 1, 2$$

$$R_s[m] - h[1]R_s[m-1] - h[2]R_s[m-2] = 0, \quad m = 1, 2$$



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$$\begin{cases} h[1]R_s[0] + h[2]R_s[-1] = R_s[1] \\ h[1]R_s[1] + h[2]R_s[0] = R_s[2] \end{cases}$$

$$\begin{bmatrix} R_s[0] & R_s[-1] \\ R_s[1] & R_s[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix}$$

$$R_s[0] = \frac{\sigma_l^2}{1-\rho^2}, \quad R_s[1] = R_s[-1] = \frac{\sigma_l^2}{1-\rho^2}\rho, \quad R_s[2] = \frac{\sigma_l^2}{1-\rho^2}\rho^2$$

$$\frac{\sigma_l^2}{1-\rho^2} \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{\sigma_l^2}{1-\rho^2} \begin{bmatrix} \rho \\ \rho^2 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}^{-1} \begin{bmatrix} \rho \\ \rho^2 \end{bmatrix} = \frac{\rho}{1-\rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \rho \end{bmatrix} = \begin{bmatrix} \rho \\ 0 \end{bmatrix}$$

$$\hat{S}[n] = \rho S[n-1]$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])S[n]\} \\ &= R_s[0] - h[1]R_s[1] - h[2]R_s[2] \\ &= \frac{\sigma_l^2}{1-\rho^2} - \rho \cdot \frac{\sigma_l^2}{1-\rho^2}\rho - 0 \cdot \frac{\sigma_l^2}{1-\rho^2}\rho^2 = \sigma_l^2 \end{aligned}$$

When $\rho = 0.9$ and $\sigma_l^2 = 2$:

$$R_s[0] = 10.5263, \quad R_s[1] = R_s[-1] = 9.4737, \quad R_s[2] = 8.5263$$

$$\begin{bmatrix} 10.5263 & 9.4737 \\ 9.4737 & 10.5263 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 9.4737 \\ 8.5263 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 0.9 \\ 0 \end{bmatrix}$$



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$$\hat{S}[n] = 0.9S[n-1], \quad \sigma_\varepsilon^2 = \sigma_I^2 = 2$$

$$(3) \quad \hat{S}[n] = h[1]X[n-1] + h[2]X[n-2] + b$$

Since $S[n]$ and $X[n]$ are zero mean processes we have that: $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.

$$E\{\varepsilon[n]X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - \hat{S}[n])X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])X[n-m]\} = 0, \quad m = 1, 2$$

$$R_{SX}[-m] - h[1]R_X[m-1] - h[2]R_X[m-2] = 0, \quad m = 1, 2$$

$$\begin{cases} h[1]R_X[0] + h[2]R_X[-1] = R_{SX}[-1] = R_S[1] \\ h[1]R_X[1] + h[2]R_X[0] = R_{SX}[-2] = R_S[2] \end{cases}$$

$$\begin{bmatrix} R_X[0] & R_X[-1] \\ R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}$$

$$R_X[m] = R_S[m] + R_W[m] = R_S[m] + \sigma_w^2 \delta[m], \quad \gamma = \frac{\sigma_s^2}{\sigma_w^2}$$

$$R_X[0] = \sigma_s^2 + \sigma_w^2 = \sigma_w^2(1 + \gamma), \quad R_X[1] = R_S[1] = \sigma_s^2 \rho = \sigma_w^2 \gamma \rho, \quad R_S[2] = \sigma_s^2 \rho^2 = \sigma_w^2 \gamma \rho^2$$

$$\begin{bmatrix} \sigma_w^2(1 + \gamma) & \sigma_w^2 \gamma \rho \\ \sigma_w^2 \gamma \rho & \sigma_w^2(1 + \gamma) \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} \sigma_w^2 \gamma \rho \\ \sigma_w^2 \gamma \rho^2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 + \gamma & \gamma \rho \\ \gamma \rho & 1 + \gamma \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \gamma \rho \begin{bmatrix} 1 \\ \rho \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \gamma \rho \begin{bmatrix} 1 + \gamma & \gamma \rho \\ \gamma \rho & 1 + \gamma \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ \rho \end{bmatrix} = \frac{\gamma \rho}{(1 + \gamma)^2 - \gamma^2 \rho^2} \begin{bmatrix} 1 + \gamma & -\gamma \rho \\ -\gamma \rho & 1 + \gamma \end{bmatrix} \begin{bmatrix} 1 \\ \rho \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{\gamma \rho}{(1 + \gamma)^2 - \gamma^2 \rho^2} \begin{bmatrix} 1 + \gamma - \gamma \rho^2 \\ \rho \end{bmatrix} \Rightarrow h[1] = \frac{\gamma \rho (1 + \gamma - \gamma \rho^2)}{(1 + \gamma)^2 - \gamma^2 \rho^2}, \quad h[2] = \frac{\gamma \rho^2}{(1 + \gamma)^2 - \gamma^2 \rho^2}$$



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$$\begin{aligned}\sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} = E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])S[n]\} \\ &= R_s[0] - h[1]R_{sx}[-1] - h[2]R_{sx}[-2] = R_s[0] - h[1]R_s[1] - h[2]R_s[2] \\ &= \sigma_s^2(1 - \rho h[1] - \rho^2 h[2])\end{aligned}$$

When $\rho = 0.9$, $\sigma_l^2 = 2$, and $\sigma_w^2 = 5$:

$$R_x[0] = 15.5263, \quad R_x[1] = R_s[1] = 9.4737, \quad R_s[2] = 7.6737$$

$$\begin{bmatrix} 15.5263 & 9.4737 \\ 9.4737 & 15.5263 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 9.4737 \\ 8.5263 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 15.5263 & 9.4737 \\ 9.4737 & 15.5263 \end{bmatrix}^{-1} \begin{bmatrix} 9.4737 \\ 8.5263 \end{bmatrix} = \begin{bmatrix} 0.4383 \\ 0.2817 \end{bmatrix}$$

The same results could have been obtained by inserting $\rho = 0.9$ and $\gamma = 10.5263/5 = 2.1053$ in:

$$h[1] = \frac{\gamma\rho(1 + \gamma - \gamma\rho^2)}{(1 + \gamma)^2 - \gamma^2\rho^2} = 0.4383, \quad h[2] = \frac{\gamma\rho^2}{(1 + \gamma)^2 - \gamma^2\rho^2} = 0.2817$$

$$\hat{S}[n] = 0.4383 \cdot X[n-1] + 0.2817 \cdot X[n-2]$$

$$\sigma_\varepsilon^2 = \sigma_s^2(1 - \rho h[1] - \rho^2 h[2]) = 3.9721 > 2$$

Obviously, in the presence of AWGN the MSE increases.